3	12	-1	-12
4	17	0	0
5	14	1	14
6	10	2	20
7	6	3	18
	$\Sigma f_i=73$		$\Sigma f_i d_i=7$

$$\overline{x} = A + \frac{\sum_{i=1}^{n} f_i d_i}{N}$$

$$= 4 + \frac{7}{73}$$

$$= 4.09$$

The mean number of mistakes = 4.09

(c) Mean for Continuous Grouped data:

For the computation of A.M for the continuous grouped data, we can use direct method or short cut method.

(i) Direct Method:

The formula is

$$\overline{x} = \frac{\sum_{i=1}^{n} f_i x_i}{N}, \quad x_i \text{ is the midpoint of the class interval}$$

(ii) Short cut method

$$\overline{x} = A + \frac{\sum_{i=1}^{n} f_i d_i}{N} \times C$$

$$d = \frac{x_i - A}{c}$$

where A - any arbitrary value

c - width of the class interval

x_i is the midpoint of the class interval.

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